Random Multispace Quantization as an Analytic Mechanism for BioHashing of Biometric and Random Identity Inputs

Andrew B.J. Teoh, *Member*, *IEEE*, Alwyn Goh, and David C.L. Ngo, *Member*, *IEEE*

Abstract—Biometric analysis for identity verification is becoming a widespread reality. Such implementations necessitate large-scale capture and storage of biometric data, which raises serious issues in terms of data privacy and (if such data is compromised) identity theft. These problems stem from the essential permanence of biometric data, which (unlike secret passwords or physical tokens) cannot be refreshed or reissued if compromised. Our previously presented biometric-hash framework prescribes the integration of external (password or token-derived) randomness with user-specific biometrics, resulting in bitstring outputs with security characteristics (i.e., noninvertibility) comparable to cryptographic ciphers or hashes. The resultant BioHashes are hence cancellable, i.e., straightforwardly revoked and reissued (via refreshed password or reissued token) if compromised. BioHashing furthermore enhances recognition effectiveness, which is explained in this paper as arising from the Random Multispace Quantization (RMQ) of biometric and external random inputs.

Index Terms—Cancellable biometrics, BioHashing, random multispace quantization, face recognition.

1 Introduction

Cancellable biometrics, a concept introduced by Bolle et al. [1], refers to the intentional and systematically repeatable distortion of biometric data in order to protect sensitive user-specific features. As elaborated from the stipulations of Maltoni et al. [2], the principal objectives of such a cancellable biometric template are:

1. Diversity: The same cancellable template cannot be used in two different applications.
2. Reusability: Straightforward revocation and reissue in the event of compromise.
3. Noninvertibility of template computation to prevent recovery of biometric and external factors.
4. Performance: The cancellable biometric template should not deteriorate the recognition performance.

This paper presents a method that conforms to these cancellability criteria. The outline of the paper is as follows: Section 2 contains a literature survey on related research. Section 3 outlines the RMQ formulation, while Section 4 gives a brief introduction to FDA—the feature extractor for face data used in this paper. Section 5 elaborates on the RMQ formulation in detail and Section 6 presents the experimental results and the discussion, followed by the concluding remarks in Section 7.

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2 Related Research

Several cancellable biometric formulations have been proposed in the literature. Davida et al. [3] made the first attempt toward this direction. Davida et al. outlined cryptographic signature verifcation of iris data without stored references. This is accomplished via open token-based storage of user-specifc error correction codes to rectify offsets in the test data, thereby allowing verification of the corrected biometric and recovery of iris data via analysis of these codes. To a certain extent, the scheme may preserve user privacy as the biometric template was noninvertible. However, neither of the issues of reusability or practical work has been addressed in this scheme. Juels and Wattenberg [5] and Juels and Sudan [6] generalized and extended the Davida et al. scheme, resulting in demonstrably enhanced security. Clancy et al. [7] implemented the technique that proposed by Juels and Sudan [6]. In Clancy et al.'s work, a group of minutia points were extracted from an input fingerprint to bind in a locking set using a polynomial-based secret sharing scheme. Subsequently, a nonrelated chaff point was added intentionally to "shadow" the identification code to maximize the unlocking computational complexity, where the secret code could only be recovered if there is a substantial overlap between the input and testing fingerprint. The method has been theoretically proven secure in protecting the secrecy of fingerprint. Nevertheless, it is way beyond the level of practical use due to the high False Reject Rate at 20-30 percent. Besides that, the query-template alignment is also another issue to be considered [8].

Monrose et al. [9] proposed a hardened password based on keystroke dynamics. The feature descriptor was obtained based on the duration of each key and the latency between each pair of keystrokes. The security of their method was based on the computational hardness of small polynomial reconstruction. Subsequently, the same technique was also applied to voice [10]. The weaknesses of this work are low

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identification code length up to 60 bits, which is insufficient for most security applications, and unacceptably high False Rejection Rate = 20 percent.

Soutar et al. [11] proposed identification code recovery from the optical integral correlation of figerprint data and previously registered *Bioscrypts*. *Bioscrypts* results from the mixing of random number and the fingerprint data, thereby preventing recovery of the original fingerprint data with data capture uncertainties addressed via multiply-redundant majority-result table lookups. However, the scheme assumed the template and input are well aligned, which is often difficult to achieve, and there were no performance results published. Generally, the Juel et al., Monrose et al., and Soutar et al. approaches offer rigorous security as they generate the identification code from random number generator and securely bind to the user's biometrics. During the authentication stage, the code is released from the secure mixture when a genuine biometrics is presented. In other words, the biometrics act as the "key" to the identification code, in which only upon correct or sufficiently close presentation of the test biometric input will the secure code, be released. Although the code is revocable, the biometric is not. Besides that, these approaches suffer from error tolerance in the binary representation of the biometrics, which normally worsen the accuracy performance.

Bolle et al. [1] introduced an intentional distortion of a biometrics signal based on a chosen transform function. The biometrics signal was distorted in the same fashion at each presentation, that is, during enrollment and for every subsequent authentication. With this approach, every instance of enrollment can use a different transform function, thus rendering cross-matching impossible. Furthermore, if one variant of the biometrics is compromised, then the transformation can simply be changed to create a new variant for reenrollment. However, it is not an easy task to design such a function due to the characteristics of the feature vector. Generally, extracted features take different values, changing in some range depending on the type of biometrics used and the feature extractor, rather than taking precise values and, therefore, transform function has to satisfy some smoothness criteria. While providing robustness against variability of the same user's biometric data, that transformation also has to distinguish different users successfully. Tulyakov et al. [12] present a method of distorting fingerprint minutia information and performing fingerprint matching in a new domain. Since only distorted data is transmitted and stored in the server database and it is impossible to restore fingerprint minutiae locations using distorted data. Ang et al. [13] proposed a similar technique with the key-dependent transformation so that the matching can be done in the transformed domain. Yet, both transforms degrade the matching accuracy significantly in the altered domain.

Savvides et al. [14] proposed a cancellable biometrics scheme which encrypted the training images used to synthesize the correlation filter for biometrics authentication. They demonstrated that convolving the training images with any random convolution kernel prior to building the biometric filter does not change the resulting correlation output peak-to-sidelobe ratios, thus preserving the authentication performance. However, the security will be jeopardized via a deterministic deconvolution with a known random kernel.

Goh et al. [15] and Teoh et al. [16], [17] subsequently introduced the biometric-hash framework via iterative inner products between biometric vectors and token-derived random sequences. BioHashing is demonstrated to be a one-way transformation equivalent to a cryptographic cipher [20], thereby providing a high degree of protection to the biometric and external factors.

In this paper, we undertake a formal statistical analysis of the previously published biometric-hashing framework [15], [16], [17] in terms of the constituent random multispace quantization (RMQ) operations. This commences in three stages:

1. Projection of biometric to a lower-dimensioned and more discriminative feature domain using linear transformations such as Principle Component Analysis (PCA) [18] or Fisher Discrimination Analysis (FDA) [19].
2. Projection onto multiple random subspaces, the set of which is derived from the external input.
3. Quantization of these individual maps.

The resulting bitstring output is dependent on both biometric and external inputs, but irreproducible without simultaneous presentation of the two factors. This paper presents a detailed analysis of Steps 2 and 3, with particular emphasis on the statistical effects resulting in enhanced recognition effectiveness.

From the viewpoint of recognition effectiveness, the RMQ formulation enables intraclass variation of transformed face features to be preserved, while simultaneously accentuating interclass variations through remapping onto multiple random subspaces.

3 Overview of RMQ Formulation

The process is comprised of three stages: feature extraction, random multispace mapping, and, finally, quantization. In the feature extraction stage, the individual's biometric image, such as face, $\boldsymbol{i} \in \Re^N$, is linear transformed (using, for example, Fisher Discrimination Analysis (FDA)) from a spatial domain with dimension N to a lower and discriminative representation domain, $\boldsymbol{\omega} \in \Re^p$, where $N >> p$. The biometric feature vector, $\boldsymbol{\omega}$, is further mapped onto a sequence of random subspaces—as determined from an externally derived pseudorandom sequence—$\boldsymbol{R} \in \Re^{p \times m}$, where $m \leq p$. The random mix (RM) vector, $\boldsymbol{v}$, is described as follows:

$$\boldsymbol{v} = \kappa \boldsymbol{R}^T \boldsymbol{\omega}, \tag{1}$$

where κ is a constant value.

Finally, each RM vector is quantized into a binary {0, 1} outcome via application of selected threshold at $\boldsymbol{\tau}$ so that, on average, 50 percent of projections have absolute value larger than $\boldsymbol{\tau}$ and 50 percent have absolute value less than $\boldsymbol{\tau}$. The single random subspace formulation can be extended to include multiple subspaces, resulting in a 2^m bitstring, each representing a distinct RM vector. The resulting bitstring, referred to as the RMQ template, $\boldsymbol{b}_i^k$, can be described as follows:

$$\boldsymbol{b}^k = sgn\left(\kappa \boldsymbol{R}^{kT} \boldsymbol{\omega}^k - \boldsymbol{\tau}\right) \in 2^m, \;\; where\ k = 1, \ldots, g, \tag{2}$$

where $sgn(\cdot)$ is defined as the signum function and $\boldsymbol{\tau}$ is a preset threshold.

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During authentication, the resulting RMQ template is compared to some previously computed reference template associated with a particular user for closeness of match in terms of Hamming distance. The straightforward refreshment of RMQ templates via replacement of the external factor results in a different pseudorandom sequence and, hence, bitstring outcome, even with the same user biometric. Such sequences may be generated by means of secret password or serial number (associated with a physical token) as a cryptographic key or initial condition.

An attacker trying to recover the underlying biometric data has to invert the RMQ bitstring output, which is computationally infeasible due to the RMQ process being demonstrably [20]:

1. Complete: In terms of output being dependent on the entirety of biometric input.
2. Bit-independent: In terms of each quantization outcome being independent of all others.
3. Intractable: In terms of the constituent inputs being irrecoverable from the quantized outputs, which, in the case of the biometric data, is due to the impossibility of solving a system of linear equations as in (1) if $m < p$ [21].
4. High entropy outputs: In which the quantization outcomes are maximally unpredictable. This protects the biometric data to a degree equivalent to a cryptographic cipher or hash input.

We consider two scenarios that might occur in a real-world application:

1. Compromised biometric: In which fraudulent verification is attempted using only intercepted biometric data associated with the genuine user, but without the associated token or otherwise designated external factor.
2. Compromised external input: In which fraudulent verification is attempted using only the token or password associated with the genuine user, but without knowledge of the user-specific biometric.

The experimental results reported in Section 6.4 show that our method survives these anticipated attacks.

4 Feature Extraction

Fisher Discriminant Analysis (FDA) is a popular technique to maximize the ratio of the interclass scatter to the intraclass scatter. The end result is a projective transformation on face images $\boldsymbol{i} \in \Re^N$ such that images of the same user class are mapped close to each other, while images from different user classes are mapped far apart. The outcome projections are in the Fisher vector-space $\Re^{c-1}$ of reduced dimensionality $c << N$ as compared with the image bitmap-space.

Consider a set of N face images with c being the total number of classes, each class consisting of M images, $\boldsymbol{i}_1, \boldsymbol{i}_2, \ldots, \boldsymbol{i}_M$. The intraclass scatter matrix S_W and the interclass scatter matrix S_B are calculated as follows:

$$S_W = \sum_{j=1}^{c} \sum_{k=1}^{M} (\boldsymbol{i}_k^j - \bar{\boldsymbol{i}}_j)(\boldsymbol{i}_k^j - \bar{\boldsymbol{i}}_j)^T$$

and

$$S_B = \sum_{k=1}^{M} (\bar{\boldsymbol{i}}_k - \eta)(\bar{\boldsymbol{i}}_k - \eta)^T,$$

respectively, where $\bar{\boldsymbol{i}}_j = \frac{1}{M}\sum_{k=1}^{M} \boldsymbol{i}_k^j$ and $\eta = \frac{1}{c}\sum_{j=1}^{c} \bar{\boldsymbol{i}}_j$.

The generalized eigenvectors $\boldsymbol{u}$ are eigenvectors of $S_w^{-1} S_b$. The Fisher projection is given as follows:

$$\boldsymbol{\omega}_k = \boldsymbol{u}^T(\boldsymbol{i}_k - \bar{\boldsymbol{i}}), \text{ where } k = 1, \ldots, c-1. \quad (3)$$

FDA can be executed via the following two-step measure:

1. PCA [18], resulting in reduced dimensionality.
2. Fisher Discriminant Analysis in the reduced PCA eigenspace.

5 Detailed Specification of RMQ Procedure

5.1 Random Mapping of Feature Vector

In the second stage, the face feature, $\boldsymbol{\omega} \in \Re^p$, where $p = c - 1$, is remapped onto a random subspace as follows:

$$\boldsymbol{v} = \kappa \boldsymbol{R}_{pm}^T \boldsymbol{\omega}_p, \text{ where } \kappa\sqrt{\frac{p}{m}} \text{ and } m \leq p. \quad (4)$$

For such a mapping to be useful in terms of face recognition effectiveness, it is important that distances in the input feature space be preserved in the output random space. This is addressed by means of the Johnson-Lindenstrauss (JL) Lemma [24]:

J-L Lemma. *For any $0 < \varepsilon < 1$ and any integer n, let m be a positive integer such that $m \geq \frac{4ln\, n}{\varepsilon^2/2 - \varepsilon^3/3}$. Then, for any set S of $n = |S|$ data points in $\Re^p$, there is a map $f : \Re^p \to \Re^m$ such that, for all $\boldsymbol{x}, \boldsymbol{y} \in S$,*

$$(1-\varepsilon)||\boldsymbol{x} - \boldsymbol{y}||^2 \leq ||f(\boldsymbol{x}) - f(\boldsymbol{y})||^2 \leq (1+\varepsilon)||\boldsymbol{x} - \boldsymbol{y}||^2. \quad (5)$$

The lemma states that any set of n points in p-dimensional Euclidean space can be embedded into an $O(\frac{ln\, n}{\varepsilon^2})$-dimensional space such that the pairwise distance of any two points is maintained within an arbitrarily small factor. The scaling term $\sqrt{(p/m)}$ in (4) takes into account the decrease in the dimensionality of the data.

In our context, intraclass variations in $\boldsymbol{\omega}$ occur due to variations in the data capture conditions, while interclass variations serve to discriminate between different users. It is hence important to preserve in the remapped subspace the characteristics of both:

1. Robustness: Relative proximity of biometric vectors associated with the same user, captured under different conditions.
2. Discrimination: Relative separation of biometric vectors associated with different users.

The following section examines in detail the effects of random mapping matrix $\boldsymbol{R}$ and subspace dimensionality m.

5.1.1 Optimal Selection of Random Mapping Matrix

The effectiveness of random mapping depends on how it affects the preservation of pairwise distances For random basis vectors, $\boldsymbol{r}_i \in \boldsymbol{R}$ belonging to a standard normal distribution, it is sufficient to use matrices with independent

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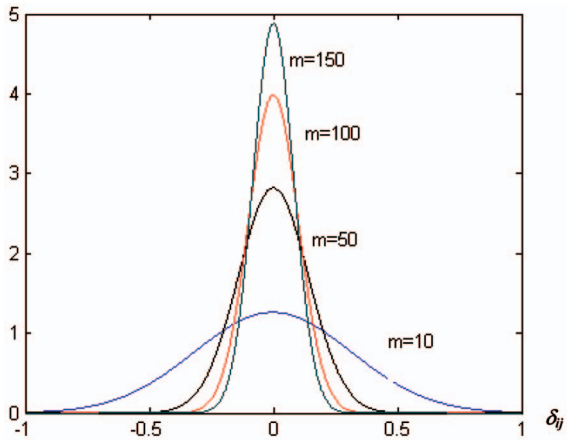

Fig. 1. δ_{ij} is distributed according to $\delta_{ij} \sim N(0, 1/m)$.

column vector entries chosen from such a distribution with bound support [25].

Let $||f(\boldsymbol{x})|| = ||f(\boldsymbol{y})|| = 1$, from (5), we have

$$\begin{aligned} ||f(\boldsymbol{x}) - f(\boldsymbol{y})||^2 &= ||f(\boldsymbol{x})||^2 + ||f(\boldsymbol{y})||^2 - 2f(\boldsymbol{x}) \cdot f(\boldsymbol{y}) \\ &= ||f(\boldsymbol{x})||^2 + ||f(\boldsymbol{y})||^2 - 2f(\boldsymbol{x})^{\mathrm{T}} f(\boldsymbol{y}) \\ &= 2(1 - f(\boldsymbol{x})^{\mathrm{T}} f(\boldsymbol{y})). \end{aligned} \tag{6}$$

Let

$$\begin{aligned} \beta \quad &= 1 - f(\boldsymbol{x})^{\mathrm{T}} f(\boldsymbol{y}) \\ &= 1 - \boldsymbol{x}^T \boldsymbol{R}_{pm}^T \boldsymbol{R}_{pm} \boldsymbol{y} \qquad 0 < \beta < 1. \end{aligned} \tag{7}$$

Since (7) is used to determine the separation between the feature vectors rather than to calculate the similarity between them, we do not need to scale the projection by $\sqrt{(p/m)}$. The matrix $\boldsymbol{R}_{pm}^T \boldsymbol{R}_{pm}$ can, without loss of generality, be decomposed as follows:

$$\boldsymbol{R}_{pm}^T \boldsymbol{R}_{pm} = \boldsymbol{I} + \delta_{ij} \;\; i \neq j, \tag{8}$$

where $\delta_{ij} = \boldsymbol{r}_i^{\mathrm{T}} \boldsymbol{r}_j, \boldsymbol{r}_i, \boldsymbol{r}_j \in \boldsymbol{R}$, and $\delta_{ii} = 0$ for $\forall i$.

If $\boldsymbol{R}_{mp}$ is orthonormal, $\delta_{ij} = 0$ for $i \neq j$ and, hence, $\boldsymbol{R}_{pm}^T \boldsymbol{R}_{pm} = \boldsymbol{I}$ and $f(\boldsymbol{x})^{\mathrm{T}} f(\boldsymbol{y}) = \boldsymbol{x}^T \boldsymbol{y}$. This indicates the pairwise distances between feature vectors are preserved after mapping to the random subspace. The orthogonal basis of $\boldsymbol{R}$ can be obtained by applying the Gram-Schmidt algorithm or its variants [26] to the row vectors and then normalizing them to unit length.

5.1.2 Random Mapping Dimensionality

In practice, the entries of δ_{ij} in (8) will not be precisely zero due to the fact that theoretically perfect orthogonality of the random signals, as required to ensure preservation of pairwise distances [27], is often difficult to obtain. We can nevertheless show that the degree of preservation of the feature topology increases with the dimension of the random subspaces, m, until a maximum is reached when the subspace dimension equals the feature dimension, i.e., $m = p$.

The effect of m can be analyzed statistically through small perturbations of $\delta_{ij} = \boldsymbol{r}_i^{\mathrm{T}} \boldsymbol{r}_j$, where $i \neq j$ and $\boldsymbol{r}_i$ and $\boldsymbol{r}_j$ are two normalized random vectors independently drawn from a standard normal distribution, $N(0, 1)$. δ_{ij} can be regarded as an estimator of the correlation coefficient between two zero-mean unit-variance random variables subject to normal distribution. Due to the Fisher transformation, δ_{ij} becomes $0.5 \ln(1 + \delta_{ij})/(1 - \delta_{ij})$, which is normally distributed with variance $1/(m - 3)$ [28]. As m becomes larger, $\boldsymbol{\sigma}_\delta^2 \approx 1/m$ and $\delta_{ij} \sim N(0, 1/m)$ (Fig. 1). In other words as m increases, the entries of δ_{ij} become smaller and, thus, $\boldsymbol{R}_{pm}^T \boldsymbol{R}_{pm} \approx \boldsymbol{I}$.

TABLE 1
Statistics Summary for Genuine Population of FDA and RM-m

	μ_g	σ_g^2
FDA	0.200	0.0030
RM-10	0.25	0.017
RM-30	0.190	0.0028
RM-60	0.200	0.0029
RM-90	0.210	0.0030

The above discussion is verified based on the face data set and the method that is described in Section 6.1. Table 1 illustrates that the gross statistical properties (as indicated by mean μ_g and variance σ_g^2) of feature vectors in FDA space and after random mapping are broadly similar, except for $m = 10$, where there is some distortion, perhaps attributable to small m. This experimental data indicates that statistical properties associated with particular users are preserved after random mapping if m is sufficiently large.

5.2 Random Multispace Quantization

The single random subspace formulation can be extended to include multiple subspaces, each representing a different individual k. Specifically, let $P = \{\boldsymbol{\omega}_j^k \in \Re^p | j = 1, \ldots, n, k = 1, \ldots, g\}$ be a set of g-by-n (individuals $\times$ images) p-dimensional feature vectors and $\Xi = \{P_1, \ldots, P_g\}$ some specified subspace partition such that $\bigcup_{i=1,\ldots,g} P_i = P$ is the union of all specified partitions and $|P_i|$ the cardinality of the set of subspaces, which is equal to the number of users g in the system for $\forall i = 1, \ldots, g$.

The multiple random subspace formulation can be defined by a set of subspaces $S = \{S_i | S_i = \kappa \boldsymbol{R}_{mp}^{i\,T} \boldsymbol{\omega}_{pm}^i, i = 1, \ldots, g\}$, where $\boldsymbol{R} \sim N(0, 1)$. For security considerations, $\boldsymbol{R}_{mp}^k$ should be mutually uncorrelated, as would be the case if formed from the output sequence of the pseudorandom generator.

In Section 3, we outline the RMQ procedure using a face data set. As previously specified, computation of random multispace mappings $\{\boldsymbol{v}^k = \kappa \boldsymbol{R}^{kT} \boldsymbol{\omega}^k | k = 1, \ldots, g\}$ follows feature vector extraction. This leads to an approximately normal distribution of $\left\{\overline{\boldsymbol{v}}^k = \frac{1}{m} \sum_{i=1}^m v_m^k | k = 1, \ldots, g\right\}$ with mean and standard deviation of $0.0006 \approx 0$ and 0.06, respectively, as illustrated in Fig. 2.

The final step, as stated earlier, is quantization on the random multispace outcomes. This paper discusses the most straightforward scheme of two-state (binary) quantization. Binarization is executed via evaluation of all $\boldsymbol{v}^k$ against some empirical threshold value τ and compilation of the collective outcomes to constitute bitstring $\boldsymbol{b}$. Specifically, let $\boldsymbol{v} = \{v_i | i = 1, \ldots, m\}$ and $\boldsymbol{b} = \{b_i | i = 1, \ldots, m\}$ such that:

$$b_i = \begin{cases} 0 & \text{if } v_i \leq \tau \\ 1 & \text{if } v_i > \tau, \end{cases} \quad \text{where } i = 1, \ldots, m. \tag{9}$$

Since the distribution of $\overline{\boldsymbol{v}}^k$ is data dependent, τ is established (at zero according to experimental data) so that half of the projective outcomes are above the threshold and

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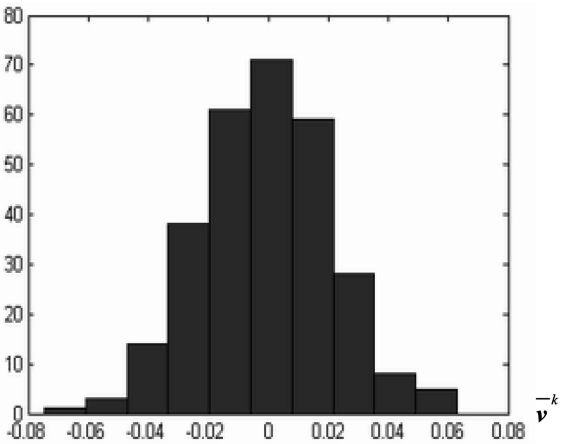

Fig. 2. Distribution of mean of random multispace mapping, $\overline{\boldsymbol{v}}^k$ for FERET data set.

the rest below. This maximizes the information content of the extracted m bits and increases the robustness of the resultant template.

5.3 Statistical Interpretation of RMQ Authentication

RMQ authentication is essentially the *failure of a test of statistical independence*, similar to the Daugman IrisCode [29] prescription. This test is statistically inclined to succeed when RMQ templates computed from different individuals are compared and correspondingly to fail when RMQ templates of the same individual are compared. The measure of bitwise disagreements corresponding to the number of subspaces in which there are substantive vector differences is straightforwardly obtained via the XOR operation:

$$d_{HD} = \boldsymbol{b}^i \oplus \boldsymbol{b}^j, \text{ where } i \neq j. \tag{10}$$

Recall from the previous section that the g RMQ templates, each representing a different user, are uncorrelated. Each of the g templates is the outcome of a Bernoulli trial, therefore collectively contributing to an imposter distribution, as in Fig. 3, which can be interpreted as a binomial distribution having mean HD $d_{HD} = 0.5m$ and with degree of freedom $\gamma = m$ [30]. Binomial distributions have functional form:

$$f(x) = \frac{\gamma!}{\lambda!(\gamma-\lambda)!}\pi^{\lambda}(1-\pi)^{(\gamma-\lambda)},$$
$$\text{with expectation } \pi \text{ and standard deviation } \sqrt{\frac{\pi(1-\pi)}{m}}, \tag{11}$$

where $x = \lambda/\gamma$ is the outcome fraction of γ Bernoulli trials, and $\pi = 0.5$. In our case, x is the HD, the fraction of bits that happen to agree when templates from two different individuals are compared. This implies that the imposter distribution will center roughly about 0.5 and, as m increases, so will the standard deviation decrease to yield a steeper slope, as shown in Fig. 3.

However, in a real-world scenario, truly random RMQ outcomes are not possible due to internal correlation between the biometrics features. Bernoulli trials that are correlated remain binomially distributed but, with a reduction in γ, the effective number of tosses [30] is:

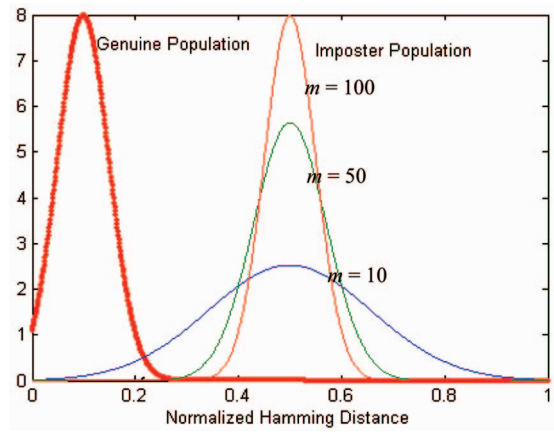

Fig. 3. Genuine-imposter population distribution for RMQ authentication.

$$\gamma_c = \mu_i(1-\mu_i)/\sigma_i^2. \tag{12}$$

The reduction rate can be measured by:

$$\eta = \frac{\gamma - \gamma_c}{\gamma}100\%, \tag{13}$$

where μ_i and σ_i^2 are the empirical mean and variance of the imposter distribution, respectively.

As illustrated in Fig. 3, the genuine user and imposter distributions tend to separate better as m increases. The imposter distribution is shifted to the right and centered at 0.5, indicating a high level of randomization in the distribution. On the other hand, the genuine distribution is preserved when m is large. The clear separation indicates that our approach results in dramatically reduced error rates in comparison to user versus imposter classification based on measurement of continuously valued differences in feature space.

The RMQ template, is hence, effective, yet computationally inexpensive. Establishment of random matrix $\boldsymbol{R}_{mp}$ and $\{\boldsymbol{\omega}_i \in \Re^p | i = 1, \ldots, n\}$ for the multiple subspace mappings requires $O(pmn)$, with $m \leq p$, for each user class. Note, each class also requires a distinct learning stage corresponding to the feature vector representation, the complexity of which is linear in g.

6 Experiments and Discussions

6.1 Data Sets and Experiment Settings

We test the efficacy of the RMQ framework for face recognition on the FERET data set [31]. FERET contains face images associated with 1,196 individuals; however, a substantial number of these (i.e., left and right-profile views) are unsuitable for our experiments. For this paper, we selected a subset of 400 users (each having four essentially frontal images with minor variations in pose (i.e., within ± 15), scale and illumination); 400 images of 100 subjects are used for training and the remaining 1,200 images of 300 subjects for testing purposes. For the FAR test (imposter distribution), the first image of each subject in the testing set is matched against the first image of all other subjects, yielding a total of $299 \times 300 \times 1/2 = 44,850$ computations. This process is repeated four times for each of the four images per user, yielding a total of $44,850 \times 4 = 179,400$ evaluations. For the FRR test (genuine distribution), each image of each subject is matched against all other images of the same subject, yielding a total of 6 (attempts of per user) $\times 300 = 1,800$ evaluations.

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Fig. 4. Examples of normalized FERET images used in our experiments.

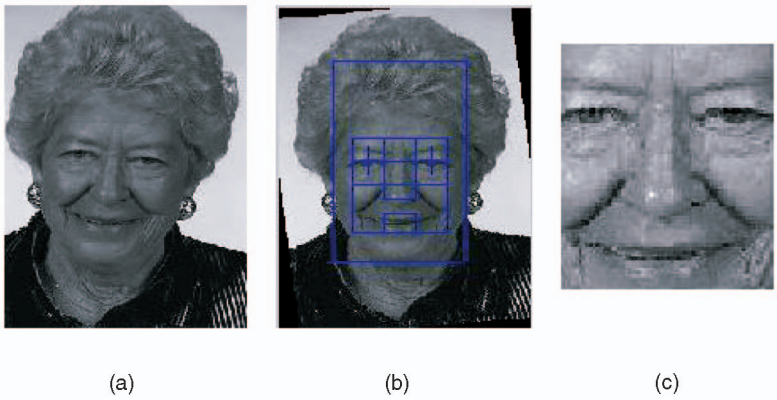

Fig. 5. Dynamic symmetry face normalization: (a) original image, (b) dynamic symmetry framing and rotation correction, and (c) cropped image at 73×61 pixels.

As the focus of the paper is on the efficacy with respect to user versus imposter classification, it is important to incorporate preprocessing mechanisms that contribute to recognition robustness. To this end, we performed geometrical normalization in order to establish correspondence between face images to be compared. The procedure is based on automatic location of the eye positions, from which various parameters (i.e., rotation, scaling, and translation) are used to extract the central part of the face from the original data set image. The examples of normalized FERET images are shown in Fig. 4.

For the following experiment, all faces were subjected to DynaFace geometric normalization, as illustrated in Fig. 5 [32]. This geometric model frames a frontal face using a golden rectangle, bisected horizontally by a line joining the centers of both eyes. The essential proportions are shown in a single geometrical figure superimposed on the illustrated frontal view. Within this rectangle, there are four main horizontal divisions:

1. the line above the eye-to-eye region,
2. the line above the nose region,
3. the line above the mouth region, and
4. the line above the chin region.

This divides the face into the following areas:

1. forehead,
2. eyes,
3. nose,
4. mouth, and
5. chin.

With regard to the evaluation of the separation between the user and imposter distributions, we used a performance indicator developed by Daugman [29]:

$$d = \frac{|\mu_g - \mu_i|}{\sqrt{\frac{1}{2}(\sigma_g^2 + \sigma_i^2)}}, \tag{14}$$

where μ_g and μ_i are the respective means of the genuine and imposter populations and σ_g^2 and σ_i^2 are the respective standard deviations. Note this figure of merit will be high if there is a large separation between the two distributions. We also evaluated our method in terms of False Acceptance Rate (FAR), False Rejection Rate (FRR), and Equal Error Rates (EER). The analysis in the following sections uses the following abbreviations:

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TABLE 2
Statistics Measurements Summary for FDA and RMQ-m, where $m = 10$, 30, 60, and 90

	m (γ)	D	μ_g	μ_i	σ_g^2	σ_i^2	γ_c	η
FDA	-	1.05	0.20	0.41	0.0300	0.0481	-	-
RMQ-m	10	1.11	0.25	0.47	0.0421	0.0337	7.40	26.00
	30	1.78	0.21	0.50	0.0391	0.0138	18.07	39.76
	60	2.54	0.19	0.50	0.0240	0.0047	52.80	12.00
	90	2.64	0.20	0.50	0.0230	0.0029	86.21	4.21

- FDA to indicate Fisher Discrimination Analysis (FDA) with upper bound, $c - 1 = 99$.
- RMQ-m to indicate FDA followed by RMQ, with the output bitlength $m(\leq c - 1)$ and $m = 10$, 30, 60, and 90.

For quantitative evaluation of dissimilarity, we use β as defined in (7) for FDA, and d_{HD}/m in (10) for RMQ-m.

6.2 RMQ Performance Evaluation

We tested various RMQ configurations, i.e., various output bitlengths m for fixed FDA feature length of $p = 90$ with $m \leq p$ for recognition performance on the FERET database. In terms of the separability measure of (14), FDA results in $d = 1.05$, while RMQ-m (Table 2) for $m = 10$, 30, 60, and 90, respectively, results in increased separations, indicating enhanced recognition performance.

Table 3 comparing the various error (i.e., FAR, FRR, and EER) rates confirms the tendency for enhanced recognition performance in response to increases in m, with an EER of 0.002 percent for $m = 90$. This is a major improvement over FDA recognition.

Note the stipulated design criteria of representation efficiency with compact bitstrings as opposed to relatively bulky floating-point vectors does not jeopardize recognition effectiveness. RMQ also resolves the issue of FAR versus FRR trade-offs at the relatively high EERs of FDA recognition. For example, to satisfy a requirement of near-zero FAR, a system based on RMQ-90 can be operated at FRR = 0.43 percent, while a corresponding system based on FDA would have to operate at a high FRR of 42.71 percent.

The separation of the genuine and imposter distributions can be qualitatively described as a combination of decreased μ_g, σ_g^2, and σ_i^2 in response to increased m, as indicated in Table 2. The occurrence of such a trend can be predicted from (11), i.e., $\sigma_i^2 = \frac{0.25}{m}$. Note also that, for the imposter all RMQ-m distributions peak at a normalized Hamming distance 0.5, denoting bitwise disagreement of half the quantized outcomes. This constitutes strong support for the proposition in Section 5.3 that an impersonation attempt is essentially a Bernoulli trial with the predicted proportion of bitwise disagreement between different RMQ templates at $\pi = 0.5$.

TABLE 3
Performance Evaluation for FDA and RMQ-m, where $m = 10$, 30, 60, and 90

	M	FAR(%)	FRR(%)	EER(%)	FRR(%)(FAR=0%)
FDA	-	5.92	3.13	4.52	42.71
RMQ-m	10	6.94	7.23	7.09	57.15
	30	3.21	2.12	2.66	17.50
	60	2.29	1.04	1.67	5.19
	90	0.002	0.001	0.002	0.43

Unfortunately, there exists some finite interuser correlation in FDA feature space, resulting in a reduced degree of freedom from $\gamma\ (= m)$ to $\gamma_c\ (= \mu_i(1 - \mu_i)/\sigma_i^2)$ pertaining to the statistical validity of the random Bernoulli model. Fortunately, this reduction η is attenuated for large m (Table 2). Note that $\gamma = \gamma_c = m$ for vanishing η, indicating a maximum degree of decorrelation among the individual RMQ outcomes. This supplements the statistical characteristics of the imposter distribution with presumed (and observed) means at $\pi = 0.5$ and standard deviation of $\sigma_i = 0.5/\sqrt{m}$ (11), which decreases with increasing m. The implication is that large m (subject to a maximum of $m = p$) ensures a clear separation between the genuine and imposter distributions and, hence, zero error rate. This can also be illustrated through the genuine-imposter distributions for FDA and RMQ-90, as shown in Fig. 6.

6.3 Effect of Feature Extraction Methodology on RMQ Process

In this section, we study the effect of different feature extraction methods, specifically, FDA and PCA [18] on the follow-on RMQ postprocessing. The experimental results for various RMQ configurations with output bitlength $m = p$ at the maximum setting of feature space dimension are presented in Table 4.

We found the performance of FDA to be consistently better than that of PCA, which is consistent with the results previously reported in [19], and that the $\mathrm{FDA} + \mathrm{RMQ}$-$m$ extensions uniformly out-perform $\mathrm{PCA} + \mathrm{RMQ}$-$m$ for corresponding m. Note also that $\mathrm{FDA} + \mathrm{RMQ}$-$p$ and $\mathrm{PCA} + \mathrm{RMQ}$-$p$ analysis yields consistently lower EERs than the corresponding FDA-p and PCA-p feature vector analysis. This confirms the efficacy of the proposed RMQ processing. Of particular interest is the leveling off of PCA-p and $\mathrm{PCA} + \mathrm{RMQ}$-$p$ recognition performance for high m dimensionalities with no further improvement beyond $p = 60$. This can be attributed to the inherent overdescriptiveness of high dimension in PCA feature vectors, resulting in essentially random noise in the high-order projective components. FDA is therefore a better choice as the feature space descriptor to be combined with RMQ postprocessing.

6.4 RMQ Analysis

Application of RMQ for identity verification presumes that each user is associated with an external digital input (i.e.,

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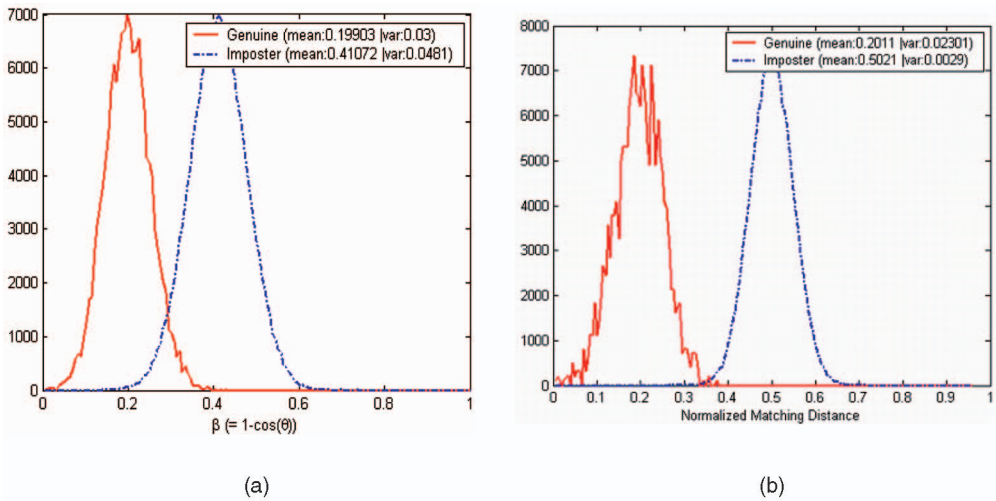

Fig. 6. Genuine and Imposter distribution for (a) FDA and (b) RMQ-90.

secret password or physical token) from which a unique random map sequence is derived. This raises the possibility of two identity theft scenarios:

1. Compromised biometric, in which an imposter possesses intercepted biometric data of sufficiently high quality to be considered authentic under feature vector analysis.
2. Compromised external input, in which an imposter has access to password or token and can hence reproduce the user-specific map sequence.

Scenario 1: Compromised Biometric. Each user subjected to this scenario has four faceviews, each of which is combined with a different external input, resulting in four RMQ-m (with $m = 90$). Bitstrings in each (of 300) user class are then compared with others in the same user class, resulting in 1,800 (from six comparisons per user) normalized Hamming distance measurements for the Pseudoimposter 1[1] distribution. This distribution (Table 5) is centerd at a mean of 0.48 with a variance of 0.003. Note the imposter distribution's mean and the variance (0.50 and 0.003) of RMQ-90 are almost identical to Pseudoimposter 1, illustrating that the net effect is essentially equivalent to projection of the compromised feature vector onto multiple random subspaces.

TABLE 4
Evaluation Performance for PCA, FDA, $\mathrm{PCA} + \mathrm{RMQ}$-$m$, and $\mathrm{FDA} + \mathrm{RMQ}$-$m$ with $m = p$

Setting		FAR(%)	FRR(%)	EER(%)
PCA	p=10	14.80	12.26	13.53
	p=30	12.89	12.13	12.51
	p=60	8.60	8.67	8.63
	p=90	8.44	8.49	8.46
FDA	p=10	13.53	10.26	11.89
	p=30	12.89	8.09	10.49
	p=60	7.72	7.73	7.73
	p=90	5.92	3.13	4.52
PCA+ RMQ-m	m=10	7.41	7.51	7.46
	m=30	4.87	4.70	4.79
	m=60	1.34	1.48	1.41
	m=90	1.31	1.56	1.44
FDA+ RMQ-m	m=10	7.11	6.81	6.96
	m=30	2.20	2.95	2.58
	m=60	0.50	0.96	0.73
	m=90	0.002	0.001	0.002

Scenario 2: Compromised External Input. Here, the same external input is used to generate a common map sequence for all 300 subjects, each resulting in an RMQ-m (with $m = 90$) which is then compared to all others. This procedure is repeated for each of the four user-specific faceviews, resulting in a total of 179,400 (from $44,850 \times 4$) Hamming distance measurements for the Pseudoimposter 2[2] distribution. This distribution (Table 5) is centerd at a mean of 0.43 with a variance of 0.048. It is interesting to observe that the Pseudoimposter-2 distribution's mean and the variance are close to the mean and variance (0.41 and 0.048) of FDA. This demonstrates that the compromise of the external input in and of itself is not particularly useful due to the discriminative effect of the interuser feature space separation and consequent preservation of these distances under the featured RMQ prescription. In other words, the Pseudoimposter 2 reverts to its original state (FDA in this context) or becomes poorer due to the quantization process. Fig. 7 depicts the comparative performance of FDA, RMQ-90, and its two compromised scenarios in Receiver Operating Curve (ROC). Note that Scenario 1 performance is closed to RMQ-90, but Scenario 2 is slightly poorer compare to FDA due to the quantization effect. This concludes that the RMQ survives these anticipated attacks.

6.5 Discussions and Comparisons

One may argue that the external digital input (i.e., secret password or physical token) may overpower the biometric in the RMQ formulation and, thus, contribute such a high verification performance in the normal RMQ and Scenario 1

1. Pseudoimposter 1 refers to imposter distribution that was generated from Scenario 1.
2. Pseudoimposter 2 refers to imposter distribution that was generated from Scenario 2.

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TABLE 5
Statistics of RMQ Analysis in Scenario 1 and 2

	Mean	Variance
*RMQ-90 (Imposter)	0.50	0.003
*FDA (Imposter)	0.41	0.048
Scenario 1 (Pseudo-Imposter 1)	0.48	0.003
Scenario 2 (Pseudo-Imposter 2)	0.43	0.048

** Reference (refer to Table 2).*

that resulting biometrics role is nullified. However, we contend that both components (external input + biometrics) play equally important roles in RMQ. For instance, if the external digital input overtakes biometric, the most apparent effect is the zero mean and standard deviation occurrences in the genuine distribution. However, this does not agree with the experimental result where genuine distribution is preserved. Furthermore, without the presence of external input, sole biometrics suffers from nonrevocable and privacy invasion issues, which are the primary concerns of the cancellable biometrics, while sole token usage is susceptible to repudiation. As for the compromised external input scenario, a straightforward solution is to use a better feature extractor as the recognition performance for this scenario is directly proportional to the quality of the feature extractor. Anyhow, due to the binary representation nature of RMQ, Error Correction Code is also worth considering.

Based on the four cancellable biometrics criteria that we highlighted in the Section 1, Table 6 compares RMQ with the prior art elaborated on in Section 2. Note that we only compare the major approaches but not their respective derivations (Section 2), which also inherited their parents' strengths and the weaknesses. In general, RMQ fulfilled all the requirements for the cancellable biometrics design, especially in the performance aspect.

7 Concluding Remarks

This paper elaborates on the biometric-hash framework by illustrating the integration of biometric and external data (derived from a secret password or physical token) in terms of a random multispace quantization (RMQ) process. This process entails transformation of raw faceviews into a low-dimension feature space representation, then subsequently remapping these user-specific feature vectors onto a sequence of random subspaces specified by a discrete external input, and, finally, quantizing these remappings to yield the RMQ biometric-hash.

The end result is an extremely powerful two-factor bio-hash which integrates biometric data with externally generated randomness in a noninvertible manner, thereby protecting sensitive biometric data in a manner equivalent to a cryptographic cipher or hash input. These biometric-hashes are, furthermore, cancellable—via straightforward revocation and then refreshment of the external random factor—thereby protecting against the interception of biometric data or even physical fabrication of the biometric feature.

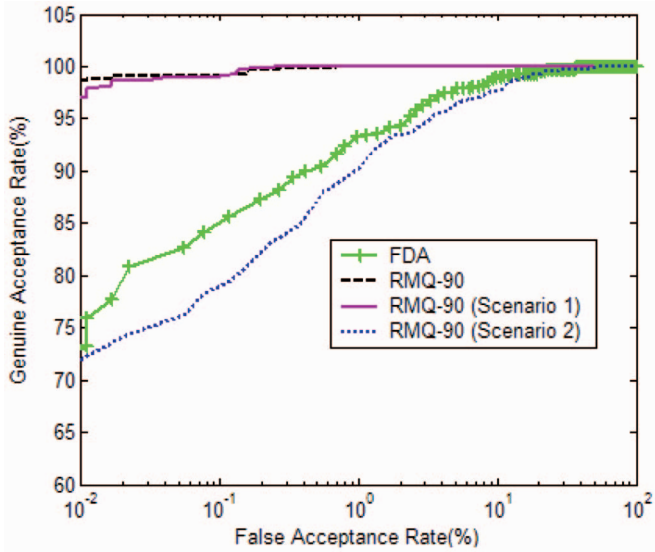

Fig. 7. ROC curve for FDA, RMQ-90, and its two compromised scenarios.

In terms of recognition performance, the proposed formulation also offers significant advantages over methods based on feature vector analysis. This can be seen from the clean separation of the genuine and the imposter populations. EERs are also reduced to near-zero level, thereby avoiding the FAR versus FRR trade-offs, which are a structural weakness of feature vector analysis. This is accomplished through the RMQ effect of preserving intrauser variations, while amplifying interuser variations via mapping onto uncorrelated random subspace sequences. Recognition performance improves with the output bitlength m, up to the maximum of $m = p$, i.e., the feature space dimension. Output bitlengths are also commensurate with the desired level of security (equivalent to cryptographic systems) against brute-force random-guessing attacks. Large m furthermore suppresses interclass correlations in the bitstring outcomes as can be seen from the predicted standard deviation of $0.5/\sqrt{m}$ for the imposter distribution, resulting in more pronounced shifting away from the genuine distribution. There is an important proviso, namely, that recognition effectiveness also depends on the quality of the feature extractor with the high-dimension vector spaces preferred in this context.

TABLE 6
A Summary of the Comparative Merits of Various Cancellable Biometrics Techniques

Methods	Diversity	Reusability	Non-invertibility	Performance
Davida et al. [3]	No	No	Yes	No Report
Bolle et al.[1]	Perhaps	Yes	Yes	No Report
Soutar et al. [11]	Yes	Yes	Yes	No Report
Monrose et al. [9][10]	Yes	Yes	Yes	Poor
Juels & Sudan [5][6]	Yes	Yes	Yes	Poor
Saviddes [14]	Yes	Yes	No	Moderate*
RMQ	Yes	Yes	Yes	Good

** Depending on the type of correlation filter performance.*

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The methodology presented is, hence, a substantive improvement over recognition based purely on feature extraction. Note that RMQ biometric-hashing is straightforwardly applicable on other biometrics forms, i.e., fingerprint, iris, and speech data. The other promising research area is the further stabilization of the bitstring outputs via error correction techniques, i.e., algebraic codes or modular polynomial interpolation. This enables RMQ biometric-hashes to be used as cryptographic keys, thereby addressing application scenarios beyond identity verification.

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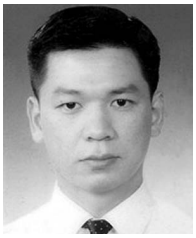

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